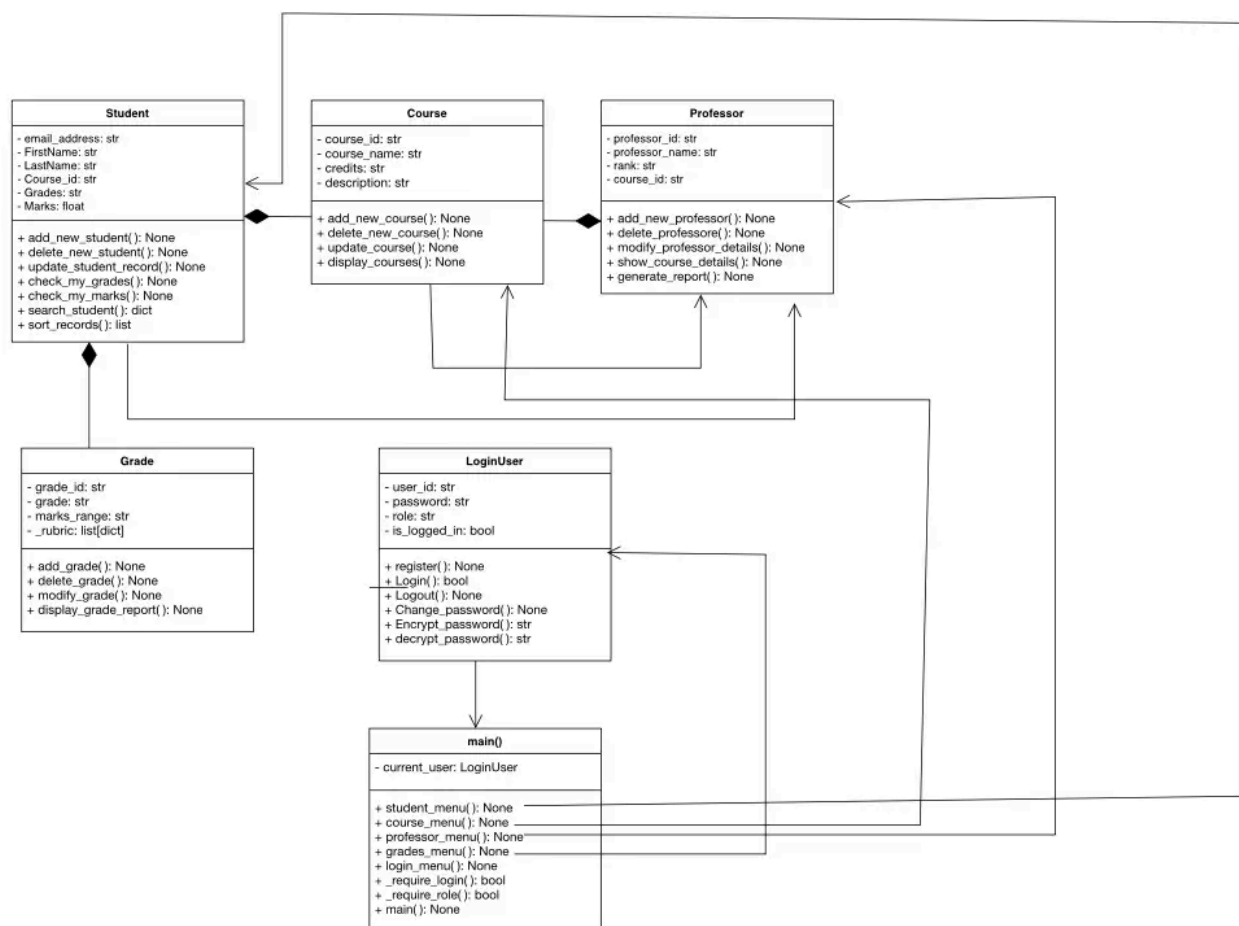


LAB 1

This lab report documents the implementation and testing of the "Check My Grade" system, a Python-based academic management application. The system supports three user roles — student, professor, and admin — and provides comprehensive functionality for managing users, courses, professors, students, and grades. Each section below demonstrates a specific feature of the system through terminal screenshots, showcasing both successful operations and error handling.



Part 1: User Registration

Part 1 covers the user authentication system, which serves as the entry point for all system interactions. The login module supports new user registration with role

assignment, secure login and logout, and password management. Passwords are validated against complexity requirements at registration and stored in encrypted form. The following screenshots demonstrate each authentication scenario, including successful flows and failure cases such as incorrect credentials and weak passwords.

1.1. User Registration

```
--- LOGIN MENU (Not logged in) ---
1. Register
2. Login
3. Logout
4. Change Password
5. Back
Select option: 1
User ID (email): jaypas@sjsu.edu
Password: R3load@24680
Role (student / professor / admin): student
User 'jaypas@sjsu.edu' registered as 'student' (password encrypted).
```

I registered a new user by selecting option 1 from the Login Menu, entering the email `jaypas@sjsu.edu`, password `R3load@24680`, and role as "student." The system stores the user's credentials by mapping the email as a unique identifier and encrypting the password before saving it, confirming successful registration.

1.2 Successful Login

```
--- LOGIN MENU (Not logged in) ---
1. Register
2. Login
3. Logout
4. Change Password
5. Back
Select option: 2
User ID: jaypas@sjsu.edu
Password: R3load@24680
Login successful. Welcome, jaypas@sjsu.edu [role: student]
```

I logged in using option 2 with the correct credentials. The system looks up the entered email, decrypts the stored password, and compares it against the input — welcoming me as `jaypas@sjsu.edu` with the role of "student" upon a successful match

1.3 Failed Login — Wrong Password

```

--- LOGIN MENU (Not logged in) ---
1. Register
2. Login
3. Logout
4. Change Password
5. Back
Select option: 2
User ID: jaypas@sjsu.edu
Password: R3load@246
Login failed: invalid user ID or password.

```

I intentionally entered a wrong password to test error handling. The system performs the same credential lookup and comparison but returns a generic "Login failed: invalid user ID or password" message without specifying which field was wrong, which is a good security practice to prevent enumeration attacks.

1.4 Weak Password Rejected at Registration

```

--- LOGIN MENU (Not logged in) ---
1. Register
2. Login
3. Logout
4. Change Password
5. Back
Select option: 1
User ID (email): jaypasnagasai@sjsu.edu
Password: r3load24
Role (student / professor / admin): student
Error: Password must be >=8 chars and include an uppercase letter, a digit, and a special character (!@#$$%^&*()-_+=).

```

I attempted to register with a weak password `r3load24`. The system runs a validation check before saving any data and rejects the input if it doesn't meet the required criteria — at least 8 characters, one uppercase letter, one digit, and one special character — displaying an error message accordingly.

1.5 Change Password

```

--- LOGIN MENU (Not logged in) ---
1. Register
2. Login
3. Logout
4. Change Password
5. Back
Select option: 4
User ID: jaypas@sjsu.edu
Old Password: R3load@24680
New Password: R3load@2468024680
Password changed successfully for 'jaypas@sjsu.edu'.

```

I selected option 4 to change the password for `jaypas@sjsu.edu`. The system first verifies the old password by decrypting the stored one, and only upon a

successful match does it encrypt and save the new password, confirming the change was successful.

1.6 Logout

```
--- LOGIN MENU (Logged in as: jaypas@sjsu.edu [role: student]) ---
1. Register
2. Login
3. Logout
4. Change Password
5. Back
Select option: 3
'jaypas@sjsu.edu' logged out successfully.
```

I selected option 3 to log out while authenticated as `jaypas@sjsu.edu` with the role of "student." The system confirmed the logout with a success message, and the menu header reverted to the "Not logged in" state in subsequent screens. This verifies that session state is properly cleared on logout.

1.7 Password Encryption/Decryption

```
Python 3.13.5 | packaged by Anaconda, Inc. | (main, Jun 12 2025, 11:23:37) [Clang 14.0.6 ] on darwin
Type "help", "copyright", "credits" or "license" for more information.
>>> import sys, os
... sys.path.insert(0, '.') # or the folder where checkmygrade.py lives
... from checkmygrade import LoginUser
...
... u = LoginUser("jaypas@sjsu.edu", "R3load@24680", "student")
...
... original = "R3load@24680"
... encrypted = u.Encrypt_password(original)
... decrypted = u.decrypt_password(encrypted)
...
... print("Original :", original)
... print("Encrypted :", encrypted)
... print("Decrypted :", decrypted)
... print("Match      :", original == decrypted)
...
Original : R3load@24680
Encrypted : EVsJDAop0XVGv1xV
Decrypted : R3load@24680
Match      : True
>>> |
```

I tested the encryption and decryption methods of the `LoginUser` class directly in the Python interpreter. I encrypted the password `R3load@24680` and then decrypted the result, confirming that the decrypted value matched the original. The `Match: True` output demonstrates that the symmetric encryption logic works correctly.

Part 2: Course Management

Part 2 demonstrates the course management functionality, which is accessible to professors and admins. This module allows authorized users to create new courses with an ID, name, credit count, and description, as well as update or remove existing ones. All course records can be displayed in a structured list. The screenshots below walk through each of these operations in sequence.

2.1 Add Courses

```
--- COURSE MENU ---
1. Display Courses
2. Add Course
3. Delete Course
4. Update Course
5. Back
Select option: 2
Course ID: DATA201
Course Name: Math For Data Science
Credits: 3
Description: The required math for getting to advanced data science
Course 'Math For Data Science' (ID: DATA201) added successfully.
```

I navigated to the Course Menu and selected option 2 to add a new course. I entered the course ID `DATA201`, named it "Math For Data Science," assigned 3 credits, and added a description. The system confirmed the course was added successfully, showing the course creation feature is functioning properly.

2.2 Update A Course

```
--- COURSE MENU ---
1. Display Courses
2. Add Course
3. Delete Course
4. Update Course
5. Back
Select option: 4
Course ID: DATA200
New Course Name (Enter to skip):
New Credits (Enter to skip): 4
New Description (Enter to skip):
Course 'DATA200' updated successfully.
```

I selected option 4 to update an existing course with ID `DATA200`. I updated the credits to 4 while leaving the name and description fields blank to skip them. The system confirmed that `DATA200` was updated successfully, demonstrating that partial updates using Enter-to-skip work as intended.

2.3 Delete A Course

```
--- COURSE MENU ---
1. Display Courses
2. Add Course
3. Delete Course
4. Update Course
5. Back
Select option: 3
Course ID to delete: ENG110
Course 'ENG110' deleted successfully.
```

I selected option 3 from the Course Menu and entered `ENG110` as the course to delete. The system confirmed the deletion with a success message, showing that course records can be removed by their course ID.

2.4 Display All Courses

```
===== CHECK MY GRADE SYSTEM =====
Status: Not logged in (go to Login Menu to sign in)

1. Student Menu
2. Course Menu      [professor / admin]
3. Professor Menu   [professor / admin]
4. Grades Menu     [professor / admin]
5. Login Menu
6. Exit

Select option: 5

--- LOGIN MENU (Not logged in) ---
1. Register
2. Login
3. Logout
4. Change Password
5. Back
Select option: 2
User ID: jaypasnagasai@sjsu.edu
Password: R3loadZ4680
Login successful. Welcome, jaypasnagasai@sjsu.edu [role: admin]

--- LOGIN MENU (Logged in as: jaypasnagasai@sjsu.edu [role: admin]) ---
1. Register
2. Login
3. Logout
4. Change Password
5. Back
Select option: 5

===== CHECK MY GRADE SYSTEM =====
Logged in as : jaypasnagasai@sjsu.edu
Role : admin

1. Student Menu
2. Course Menu      [professor / admin]
3. Professor Menu   [professor / admin]
4. Grades Menu     [professor / admin]
5. Login Menu
6. Exit

Select option: 2

--- COURSE MENU ---
1. Display Courses
2. Add Course
3. Delete Course
4. Update Course
5. Back
Select option: 1

--- Course Records ---
Course ID: DATA200 | Name: Data Science | Credits: 4 | Description: Provides insight about DS and Python
Course ID: CS101 | Name: Computer Science | Credits: 3 | Description: Introduction to programming and algorithms
Course ID: MATH301 | Name: Mathematics | Credits: 4 | Description: Advanced calculus and linear algebra
Course ID: BIO150 | Name: Biology | Credits: 3 | Description: Fundamentals of life sciences
Course ID: DATA201 | Name: Math For Data Science | Credits: 3 | Description: The required math for getting to advanced data science
```

After logging in as an admin (`jaypasnagasai@sjsu.edu`), I navigated to the Course Menu and selected option 1 to display all courses. The system listed all five stored courses with their IDs, names, credit counts, and descriptions, confirming that the display functionality retrieves all records correctly.

Part 3: Professor Management

Part 3 covers the professor management module, also restricted to professor and admin roles. This section of the system allows administrators to add new professors with their associated course assignments, modify professor details, view the courses a specific professor teaches, and remove professors from the system. The screenshots illustrate each of these operations, concluding with a full display of all professor records.

3.1 Add New Professors

```
--- PROFESSOR MENU ---
1. Display Professors
2. Add Professor
3. Delete Professor
4. Modify Professor Details
5. Show Course by Professor
6. Report by Professor
7. Back
Select option: 2
Professor ID (email): rahpat@sjsu.edu
Professor Name: Rahul Patel
Rank: Professor
Course ID: DATA201
Professor 'Rahul Patel' added successfully.
```

I selected option 2 from the Professor Menu to add a new professor. I entered the professor's email (`rahpat@sjsu.edu`), name (Rahul Patel), rank, and assigned course ID `DATA201` . The system confirmed the professor was added successfully, showing the professor creation flow works end to end.

3.2 Modify Professor Details

```

--- PROFESSOR MENU ---
1. Display Professors
2. Add Professor
3. Delete Professor
4. Modify Professor Details
5. Show Course by Professor
6. Report by Professor
7. Back
Select option: 4
Professor ID: rahpat@sjsu.edu
New Name      (Enter to skip): Rahul Amol Patel
New Rank      (Enter to skip):
New Course ID (Enter to skip):
Professor 'Rahul Amol Patel' details updated successfully.

```

I selected option 4 to update the professor with email `rahpat@sjsu.edu`. I updated the name to "Rahul Amol Patel" while skipping the rank and course ID fields. The system confirmed the update, demonstrating that professor records support partial field modification.

3.3 Show Course Details By Professor

```

--- PROFESSOR MENU ---
1. Display Professors
2. Add Professor
3. Delete Professor
4. Modify Professor Details
5. Show Course by Professor
6. Report by Professor
7. Back
Select option: 5
Professor ID: rahpat@sjsu.edu

--- Courses taught by Rahul Amol Patel ---
Course ID: DATA201 | Name: Math For Data Science | Credits: 3 | Description: The required math for getting to advanced data science

```

I selected option 5 and entered `rahpat@sjsu.edu` to see which courses that professor teaches. The system returned the course `DATA201` — "Math For Data Science" — along with its credits and description, showing that the professor-to-course lookup is correctly implemented.

3.4 Delete Professor

```

--- PROFESSOR MENU ---
1. Display Professors
2. Add Professor
3. Delete Professor
4. Modify Professor Details
5. Show Course by Professor
6. Report by Professor
7. Back
Select option: 3
Professor ID to delete: james@mycsu.edu
Professor 'james@mycsu.edu' deleted successfully.

```


I selected option 3 and entered `james@mycsu.edu` as the professor to remove. The system confirmed the deletion with a success message, verifying that professor records can be deleted by their email ID.

3.5 Display All Professors

```
===== CHECK MY GRADE SYSTEM =====
Status: Not logged in (go to Login Menu to sign in)

1. Student Menu
2. Course Menu      [professor / admin]
3. Professor Menu   [professor / admin]
4. Grades Menu     [professor / admin]
5. Login Menu
6. Exit

Select option: 5

--- LOGIN MENU (Not logged in) ---
1. Register
2. Login
3. Logout
4. Change Password
5. Back
Select option: 2
User ID: jaypasnagasai@sjsu.edu
Password: R3load@24680
Login successful. Welcome, jaypasnagasai@sjsu.edu [role: admin]

--- LOGIN MENU (Logged in as: jaypasnagasai@sjsu.edu [role: admin]) ---
1. Register
2. Login
3. Logout
4. Change Password
5. Back
Select option: 5

===== CHECK MY GRADE SYSTEM =====
Logged in as : jaypasnagasai@sjsu.edu
Role       : admin

1. Student Menu
2. Course Menu      [professor / admin]
3. Professor Menu   [professor / admin]
4. Grades Menu     [professor / admin]
5. Login Menu
6. Exit

Select option: 3

--- PROFESSOR MENU ---
1. Display Professors
2. Add Professor
3. Delete Professor
4. Modify Professor Details
5. Show Course by Professor
6. Report by Professor
7. Back
Select option: 1

--- Professor Records ---
ID: micheal@mycsu.edu | Name: Micheal John | Rank: Senior Professor | Course ID: DATA200
ID: linda@mycsu.edu | Name: Linda Carter | Rank: Associate Professor | Course ID: CS101
ID: robert@mycsu.edu | Name: Robert Hayes | Rank: Assistant Professor | Course ID: MATH301
ID: nancy@mycsu.edu | Name: Nancy Green | Rank: Professor | Course ID: BI0150
ID: rahpat@sjsu.edu | Name: Rahul Amol Patel | Rank: Professor | Course ID: DATA201
```

I navigated to option 1 in the Professor Menu to display all professors in the system. The output listed five professors with their IDs, names, ranks, and assigned course IDs, confirming that the full professor roster is retrievable and accurately reflects all current entries.

Part 4: Student Management

Part 4 presents the student management module, which is the most feature-rich section of the system. It supports adding and updating student records, checking individual grades and marks, deleting records, computing average and median marks by course or across all courses, searching for students by email with execution timing, sorting students by various fields, and generating grade reports both by individual student and by course. The following screenshots demonstrate each of these capabilities in action.

4.1 Add New Student

```
--- STUDENT MENU ---
1. Display All Records
2. Add Student
3. Delete Student
4. Update Student Record
5. Check My Grades
6. Check My Marks
7. Search Student (with timing)
8. Sort Students (with timing)
9. Average & Median Marks
10. Report by Student
11. Report by Course
12. Back
Select option: 2
Email: jaypas@sjsu.edu
First Name: Jayanth
Last Name: Pasupulati
Course ID: DATA201
Grade (e.g. A, B+): A
Marks (numeric): 95
Student 'Jayanth Pasupulati' added successfully.
```

I selected option 2 from the Student Menu and entered the details for a new student: email `jaypas@sjsu.edu`, name Jayanth Pasupulati, course `DATA201`, grade A, and marks 95. The system confirmed the student was added successfully, demonstrating that the student creation feature captures all required fields.

4.2 Update Student Record

```

--- STUDENT MENU ---
1. Display All Records
2. Add Student
3. Delete Student
4. Update Student Record
5. Check My Grades
6. Check My Marks
7. Search Student (with timing)
8. Sort Students (with timing)
9. Average & Median Marks
10. Report by Student
11. Report by Course
12. Back
Select option: 4
Email: jaypas@sjssu.edu
New First Name (Enter to skip): Jayanth Naga Sai
New Last Name (Enter to skip):
New Course ID (Enter to skip):
New Grade (Enter to skip):
New Marks (Enter to skip):
Record updated for 'Jayanth Naga Sai Pasupulati'.

```

I selected option 4 and entered the email `jaypas@sjssu.edu` to update an existing student. I changed the first name to "Jayanth Naga Sai" while skipping all other fields using Enter. The system confirmed the record was updated, showing that partial student updates are supported.

4.3 Check My Grade

```

--- STUDENT MENU ---
1. Display All Records
2. Add Student
3. Delete Student
4. Update Student Record
5. Check My Grades
6. Check My Marks
7. Search Student (with timing)
8. Sort Students (with timing)
9. Average & Median Marks
10. Report by Student
11. Report by Course
12. Back
Select option: 5
Email: jaypas@sjssu.edu
Grade for Jayanth Naga Sai Pasupulati (Course DATA201): A

```

I selected option 5 and entered `jaypas@sjssu.edu` to look up that student's grade. The system returned "Grade for Jayanth Naga Sai Pasupulati (Course DATA201): A," confirming that the grade lookup correctly maps the student to their enrolled course and stored grade.

4.4 Check My Marks

```
--- STUDENT MENU ---
1. Display All Records
2. Add Student
3. Delete Student
4. Update Student Record
5. Check My Grades
6. Check My Marks
7. Search Student (with timing)
8. Sort Students (with timing)
9. Average & Median Marks
10. Report by Student
11. Report by Course
12. Back
Select option: 6
Email: jaypas@sjsu.edu
Marks for Jayanth Naga Sai Pasupulati (Course DATA201): 95
```

I selected option 6 and entered `jaypas@sjsu.edu` to retrieve the numeric marks for that student. The system returned "Marks for Jayanth Naga Sai Pasupulati (Course DATA201): 95," showing that the marks lookup works independently from the letter grade lookup.

4.5 Delete Student Record

```
--- STUDENT MENU ---
1. Display All Records
2. Add Student
3. Delete Student
4. Update Student Record
5. Check My Grades
6. Check My Marks
7. Search Student (with timing)
8. Sort Students (with timing)
9. Average & Median Marks
10. Report by Student
11. Report by Course
12. Back
Select option: 3
Email to delete: tom@mycsu.edu
Student 'tom@mycsu.edu' deleted successfully.
```

I selected option 3 from the Student Menu and entered `tom@mycsu.edu` as the student to delete. The system confirmed the deletion, verifying that student records can be removed by their email address.

4.6 Average And Median Marks

```
--- STUDENT MENU ---
1. Display All Records
2. Add Student
3. Delete Student
4. Update Student Record
5. Check My Grades
6. Check My Marks
7. Search Student (with timing)
8. Sort Students (with timing)
9. Average & Median Marks
10. Report by Student
11. Report by Course
12. Back
Select option: 9
Course ID (Enter to skip for all): DATA200
Average marks (course 'DATA200'): 88.67
Median marks (course 'DATA200'): 88.00
```

```
--- STUDENT MENU ---
1. Display All Records
2. Add Student
3. Delete Student
4. Update Student Record
5. Check My Grades
6. Check My Marks
7. Search Student (with timing)
8. Sort Students (with timing)
9. Average & Median Marks
10. Report by Student
11. Report by Course
12. Back
Select option: 9
Course ID (Enter to skip for all):
Average marks (all courses): 87.50
Median marks (all courses): 89.50
```

I selected option 9 twice: once filtering by course `DATA200` (average 88.67, median 88.00), and once leaving the course field blank to get stats across all courses (average 87.50, median 89.50). This demonstrates that the system supports both course-specific and global statistical calculations.

4.7 Student Search

```

--- STUDENT MENU ---
1. Display All Records
2. Add Student
3. Delete Student
4. Update Student Record
5. Check My Grades
6. Check My Marks
7. Search Student (with timing)
8. Sort Students (with timing)
9. Average & Median Marks
10. Report by Student
11. Report by Course
12. Back
Select option: 7
Search by Email: alex@mycsu.edu

--- Search Result ---
Email: alex@mycsu.edu | Name: Alex Brown | Course ID: CS101 | Grade: B | Marks: 79
Time taken to search: 0.000006 seconds.

```

I selected option 7 and searched by email `alex@mycsu.edu`. The system returned Alex Brown's full record including course, grade, and marks, along with a search time of 0.000006 seconds. This confirms the search functionality works correctly and includes performance timing.

4.8 Sort Students

```

--- STUDENT MENU ---
1. Display All Records
2. Add Student
3. Delete Student
4. Update Student Record
5. Check My Grades
6. Check My Marks
7. Search Student (with timing)
8. Sort Students (with timing)
9. Average & Median Marks
10. Report by Student
11. Report by Course
12. Back
Select option: 8
Sort by: 1) Email 2) Name 3) Marks 4) Grade
Select: 4
Order - 1) Ascending 2) Descending: 2

--- Students sorted by grade (descending) ---
Email: mike@mycsu.edu | Name: Mike Williams | Course ID: MATH301 | Grade: C | Marks: 74
Email: john@mycsu.edu | Name: John Smith | Course ID: DATA200 | Grade: B | Marks: 82
Email: alex@mycsu.edu | Name: Alex Brown | Course ID: CS101 | Grade: B | Marks: 79
Email: sam@mycsu.edu | Name: Sam Carpenter | Course ID: DATA200 | Grade: A | Marks: 96
Email: emily@mycsu.edu | Name: Emily Johnson | Course ID: CS101 | Grade: A | Marks: 91
Email: sara@mycsu.edu | Name: Sara Davis | Course ID: DATA200 | Grade: A | Marks: 88
Email: lisa@mycsu.edu | Name: Lisa Wilson | Course ID: MATH301 | Grade: A | Marks: 95
Email: jaypas@sjsu.edu | Name: Jayanth Naga Sai Pasupulati | Course ID: DATA201 | Grade: A | Marks: 95
Time taken to sort: 0.000018 seconds.

```

I selected option 8 and chose to sort by grade in descending order. The system returned all students ranked from highest to lowest grade and reported a sort time

of 0.000018 seconds, demonstrating that the sorting feature supports multiple sort fields and directions with timing output.

4.9 Student Grade Report

```
--- STUDENT MENU ---
1. Display All Records
2. Add Student
3. Delete Student
4. Update Student Record
5. Check My Grades
6. Check My Marks
7. Search Student (with timing)
8. Sort Students (with timing)
9. Average & Median Marks
10. Report by Student
11. Report by Course
12. Back
Select option: 10
Student Email: lisa@mycsu.edu

--- Grade Report: Student 'lisa@mycsu.edu' ---
Name: Lisa Wilson | Course: MATH301 | Grade: A | Marks: 95
```

I selected option 10 and entered `lisa@mycsu.edu` to generate a grade report for that student. The system returned Lisa Wilson's full record showing course MATH301, grade A, and marks 95, confirming the per-student report feature pulls and formats the correct data.

4.10 Class Grade Report

```
--- STUDENT MENU ---
1. Display All Records
2. Add Student
3. Delete Student
4. Update Student Record
5. Check My Grades
6. Check My Marks
7. Search Student (with timing)
8. Sort Students (with timing)
9. Average & Median Marks
10. Report by Student
11. Report by Course
12. Back
Select option: 11
Course ID: MATH301

--- Grade Report: Course 'MATH301' ---
mike@mycsu.edu | Mike Williams | Grade: C | Marks: 74
lisa@mycsu.edu | Lisa Wilson | Grade: A | Marks: 95
--- Statistics ---
Average Marks: 84.50
Median Marks: 84.50
```

I selected option 11 and entered course ID **MATH301** to generate a report for the entire class. The system listed all enrolled students with their grades and marks, then appended statistics showing an average and median of 84.50, demonstrating that the course-level report aggregates both individual and summary data.

4.11 Display All Records

```
--- STUDENT MENU ---
1. Display All Records
2. Add Student
3. Delete Student
4. Update Student Record
5. Check My Grades
6. Check My Marks
7. Search Student (with timing)
8. Sort Students (with timing)
9. Average & Median Marks
10. Report by Student
11. Report by Course
12. Back
Select option: 1

--- Student Records ---
Email: sam@mycsu.edu | Name: Sam Carpenter | Course ID: DATA200 | Grade: A | Marks: 96
Email: john@mycsu.edu | Name: John Smith | Course ID: DATA200 | Grade: B | Marks: 82
Email: emily@mycsu.edu | Name: Emily Johnson | Course ID: CS101 | Grade: A | Marks: 91
Email: mike@mycsu.edu | Name: Mike Williams | Course ID: MATH301 | Grade: C | Marks: 74
Email: sara@mycsu.edu | Name: Sara Davis | Course ID: DATA200 | Grade: A | Marks: 88
Email: alex@mycsu.edu | Name: Alex Brown | Course ID: CS101 | Grade: B | Marks: 79
Email: lisa@mycsu.edu | Name: Lisa Wilson | Course ID: MATH301 | Grade: A | Marks: 95
Email: jaypas@sjsu.edu | Name: Jayanth Naga Sai Pasupulati | Course ID: DATA201 | Grade: A | Marks: 95
```

I selected option 1 from the Student Menu to list every student in the system. The output showed all eight students with their emails, names, course IDs, grades, and marks, confirming that the display-all function retrieves the complete and current student dataset.

Part 5: Grade Management

Part 5 covers the grade rubric management system, accessible to professors and admins. This module maintains the mapping between letter grades and their corresponding numeric mark ranges. Users can add new grade entries, modify existing ranges, delete outdated entries, and display the full grade rubric. The screenshots below illustrate each of these operations, showing the final rubric with all active grade-to-marks mappings.

5.1 Add Grade Entry


```
--- GRADES MENU ---
1. Display Grade Rubric
2. Add Grade Entry
3. Delete Grade Entry
4. Modify Grade Entry
5. Back
Select option: 2
Grade ID: 6
Grade (e.g. A, B+): A-
Marks Range (e.g. 90-100): 90-92
Grade 'A-' (ID: 6) added successfully.
```

I navigated to the Grades Menu and selected option 2 to add a new grade entry. I entered grade ID 6, grade label "A-", and a marks range of 90–92. The system confirmed the entry was added, showing that the grade rubric supports custom grade labels and numeric ranges.

5.2 Modify Grade Entry

```
--- GRADES MENU ---
1. Display Grade Rubric
2. Add Grade Entry
3. Delete Grade Entry
4. Modify Grade Entry
5. Back
Select option: 4
Grade ID: 1
New Grade (Enter to skip):
New Marks Range (Enter to skip): 93-100
Grade ID '1' updated successfully.
```

I selected option 4 and entered grade ID 1 to update its marks range to 93–100. I left the grade label blank to skip it, and the system confirmed the update was successful, demonstrating that individual grade rubric entries can be partially modified.

5.3 Delete Grade Entry

```
--- GRADES MENU ---
1. Display Grade Rubric
2. Add Grade Entry
3. Delete Grade Entry
4. Modify Grade Entry
5. Back
Select option: 3
Grade ID to delete: 5
Grade ID '5' deleted successfully.
```

I selected option 3 and entered grade ID 5 to remove it from the rubric. The system confirmed the deletion, showing that grade entries can be individually removed by their ID without affecting other entries.

5.4 Display All Grade Entries

```
--- GRADES MENU ---
1. Display Grade Rubric
2. Add Grade Entry
3. Delete Grade Entry
4. Modify Grade Entry
5. Back
Select option: 1

--- Grade Rubric ---
Grade ID: 1 | Grade: A | Marks Range: 93-100
Grade ID: 2 | Grade: B | Marks Range: 80-89
Grade ID: 3 | Grade: C | Marks Range: 70-79
Grade ID: 4 | Grade: D | Marks Range: 60-69
Grade ID: 6 | Grade: A- | Marks Range: 90-92
```

I selected option 1 to display the full grade rubric. The system listed five entries — A through D and A- — with their IDs and marks ranges, confirming that the rubric display correctly reflects all additions, modifications, and deletions made throughout the session.

Part 6: Unit Testing

```

Email: student0079@test.edu | Name: First79 Last79 | Course ID: DATA204 | Grade: B- | Marks: 99.5
Email: student0078@test.edu | Name: First78 Last78 | Course ID: DATA203 | Grade: B | Marks: 98.0
Email: student0077@test.edu | Name: First77 Last77 | Course ID: DATA202 | Grade: B+ | Marks: 98.0
Email: student0076@test.edu | Name: First76 Last76 | Course ID: DATA201 | Grade: A- | Marks: 96.5
Email: student0075@test.edu | Name: First75 Last75 | Course ID: DATA200 | Grade: A | Marks: 95.0
Email: student0074@test.edu | Name: First74 Last74 | Course ID: DATA204 | Grade: B- | Marks: 95.0
Email: student0073@test.edu | Name: First73 Last73 | Course ID: DATA203 | Grade: B | Marks: 93.5
Email: student0072@test.edu | Name: First72 Last72 | Course ID: DATA202 | Grade: B+ | Marks: 92.0
Email: student0071@test.edu | Name: First71 Last71 | Course ID: DATA201 | Grade: A- | Marks: 92.0
Email: student0070@test.edu | Name: First70 Last70 | Course ID: DATA200 | Grade: A | Marks: 90.5
Email: student0069@test.edu | Name: First69 Last69 | Course ID: DATA204 | Grade: B- | Marks: 89.0
Email: student0068@test.edu | Name: First68 Last68 | Course ID: DATA203 | Grade: B | Marks: 89.0
Email: student0067@test.edu | Name: First67 Last67 | Course ID: DATA202 | Grade: B+ | Marks: 87.5
Email: student0066@test.edu | Name: First66 Last66 | Course ID: DATA201 | Grade: A- | Marks: 86.0
Email: student0065@test.edu | Name: First65 Last65 | Course ID: DATA200 | Grade: A | Marks: 86.0
Email: student0064@test.edu | Name: First64 Last64 | Course ID: DATA204 | Grade: B- | Marks: 84.5
Email: student0063@test.edu | Name: First63 Last63 | Course ID: DATA203 | Grade: B | Marks: 83.0
Email: student0062@test.edu | Name: First62 Last62 | Course ID: DATA202 | Grade: B+ | Marks: 83.0
Email: student0061@test.edu | Name: First61 Last61 | Course ID: DATA201 | Grade: A- | Marks: 81.5
Email: student0060@test.edu | Name: First60 Last60 | Course ID: DATA200 | Grade: A | Marks: 80.0
Email: student0059@test.edu | Name: First59 Last59 | Course ID: DATA204 | Grade: B- | Marks: 80.0
Email: student0058@test.edu | Name: First58 Last58 | Course ID: DATA203 | Grade: B | Marks: 78.5
Email: student0057@test.edu | Name: First57 Last57 | Course ID: DATA202 | Grade: B+ | Marks: 77.0
Email: student0056@test.edu | Name: First56 Last56 | Course ID: DATA201 | Grade: A- | Marks: 77.0
Email: student0055@test.edu | Name: First55 Last55 | Course ID: DATA200 | Grade: A | Marks: 75.5
Email: student0054@test.edu | Name: First54 Last54 | Course ID: DATA204 | Grade: B- | Marks: 74.0
Email: student0053@test.edu | Name: First53 Last53 | Course ID: DATA203 | Grade: B | Marks: 73.0
Email: student0052@test.edu | Name: First52 Last52 | Course ID: DATA202 | Grade: B+ | Marks: 72.5
Email: student0051@test.edu | Name: First51 Last51 | Course ID: DATA201 | Grade: A- | Marks: 71.0
Email: student0050@test.edu | Name: First50 Last50 | Course ID: DATA200 | Grade: A | Marks: 71.0
Email: student0049@test.edu | Name: First49 Last49 | Course ID: DATA204 | Grade: B- | Marks: 69.5
Email: student0048@test.edu | Name: First48 Last48 | Course ID: DATA203 | Grade: B | Marks: 68.0
Email: student0047@test.edu | Name: First47 Last47 | Course ID: DATA202 | Grade: B+ | Marks: 68.0
Email: student0046@test.edu | Name: First46 Last46 | Course ID: DATA201 | Grade: A- | Marks: 66.5
Email: student0045@test.edu | Name: First45 Last45 | Course ID: DATA200 | Grade: A | Marks: 65.0
Email: student0044@test.edu | Name: First44 Last44 | Course ID: DATA204 | Grade: B- | Marks: 65.0
Email: student0043@test.edu | Name: First43 Last43 | Course ID: DATA203 | Grade: B | Marks: 63.5
Email: student0042@test.edu | Name: First42 Last42 | Course ID: DATA202 | Grade: B+ | Marks: 62.0
Email: student0041@test.edu | Name: First41 Last41 | Course ID: DATA201 | Grade: A- | Marks: 62.0
Email: student0040@test.edu | Name: First40 Last40 | Course ID: DATA200 | Grade: A | Marks: 60.5
Email: student0039@test.edu | Name: First39 Last39 | Course ID: DATA204 | Grade: B- | Marks: 99.0
Email: student0038@test.edu | Name: First38 Last38 | Course ID: DATA203 | Grade: B | Marks: 99.0
Email: student0037@test.edu | Name: First37 Last37 | Course ID: DATA202 | Grade: B+ | Marks: 97.5
Email: student0036@test.edu | Name: First36 Last36 | Course ID: DATA201 | Grade: A- | Marks: 96.0
Email: student0035@test.edu | Name: First35 Last35 | Course ID: DATA200 | Grade: A | Marks: 96.0
Email: student0034@test.edu | Name: First34 Last34 | Course ID: DATA204 | Grade: B- | Marks: 94.5
Email: student0033@test.edu | Name: First33 Last33 | Course ID: DATA203 | Grade: B | Marks: 93.0
Email: student0032@test.edu | Name: First32 Last32 | Course ID: DATA202 | Grade: B+ | Marks: 93.0
Email: student0031@test.edu | Name: First31 Last31 | Course ID: DATA201 | Grade: A- | Marks: 91.5
Email: student0030@test.edu | Name: First30 Last30 | Course ID: DATA200 | Grade: A | Marks: 90.0
Email: student0029@test.edu | Name: First29 Last29 | Course ID: DATA204 | Grade: B- | Marks: 90.0
Email: student0028@test.edu | Name: First28 Last28 | Course ID: DATA203 | Grade: B | Marks: 88.5
Email: student0027@test.edu | Name: First27 Last27 | Course ID: DATA202 | Grade: B+ | Marks: 87.0
Email: student0026@test.edu | Name: First26 Last26 | Course ID: DATA201 | Grade: A- | Marks: 87.0
Email: student0025@test.edu | Name: First25 Last25 | Course ID: DATA200 | Grade: A | Marks: 85.5
Email: student0024@test.edu | Name: First24 Last24 | Course ID: DATA204 | Grade: B- | Marks: 84.0
Email: student0023@test.edu | Name: First23 Last23 | Course ID: DATA203 | Grade: B | Marks: 84.0
Email: student0022@test.edu | Name: First22 Last22 | Course ID: DATA202 | Grade: B+ | Marks: 82.5
Email: student0021@test.edu | Name: First21 Last21 | Course ID: DATA201 | Grade: A- | Marks: 81.0
Email: student0020@test.edu | Name: First20 Last20 | Course ID: DATA200 | Grade: A | Marks: 81.0
Email: student0019@test.edu | Name: First19 Last19 | Course ID: DATA204 | Grade: B- | Marks: 79.5
Email: student0018@test.edu | Name: First18 Last18 | Course ID: DATA203 | Grade: B | Marks: 78.0
Email: student0017@test.edu | Name: First17 Last17 | Course ID: DATA202 | Grade: B+ | Marks: 78.0
Email: student0016@test.edu | Name: First16 Last16 | Course ID: DATA201 | Grade: A- | Marks: 76.5
Email: student0015@test.edu | Name: First15 Last15 | Course ID: DATA200 | Grade: A | Marks: 75.0
Email: student0014@test.edu | Name: First14 Last14 | Course ID: DATA204 | Grade: B- | Marks: 75.0
Email: student0013@test.edu | Name: First13 Last13 | Course ID: DATA203 | Grade: B | Marks: 73.5
Email: student0012@test.edu | Name: First12 Last12 | Course ID: DATA202 | Grade: B+ | Marks: 72.0
Email: student0011@test.edu | Name: First11 Last11 | Course ID: DATA201 | Grade: A- | Marks: 72.0
Email: student0010@test.edu | Name: First10 Last10 | Course ID: DATA200 | Grade: A | Marks: 70.5
Email: student0009@test.edu | Name: First9 Last9 | Course ID: DATA204 | Grade: B- | Marks: 69.0
Email: student0008@test.edu | Name: First8 Last8 | Course ID: DATA203 | Grade: B | Marks: 69.0
Email: student0007@test.edu | Name: First7 Last7 | Course ID: DATA202 | Grade: B+ | Marks: 67.5
Email: student0006@test.edu | Name: First6 Last6 | Course ID: DATA201 | Grade: A- | Marks: 66.0
Email: student0005@test.edu | Name: First5 Last5 | Course ID: DATA200 | Grade: A | Marks: 66.0
Email: student0004@test.edu | Name: First4 Last4 | Course ID: DATA204 | Grade: B- | Marks: 64.5
Email: student0003@test.edu | Name: First3 Last3 | Course ID: DATA203 | Grade: B | Marks: 63.0
Email: student0002@test.edu | Name: First2 Last2 | Course ID: DATA202 | Grade: B+ | Marks: 63.0
Email: student0001@test.edu | Name: First1 Last1 | Course ID: DATA201 | Grade: A- | Marks: 61.5
Time taken to sort: 0.000044 seconds.
ok
test_11_update_student (__main__.TestStudent.test_11_update_student) ... Record updated for 'UpdatedFirst Last1'.
--- Search Result ---
Email: student0001@test.edu | Name: UpdatedFirst Last1 | Course ID: DATA201 | Grade: A+ | Marks: 99
Time taken to search: 0.000000 seconds.
ok
test_12_update_nonexistent_student (__main__.TestStudent.test_12_update_nonexistent_student) ... Student not found.
ok
test_13_delete_student (__main__.TestStudent.test_13_delete_student) ... Student 'student1000@test.edu' deleted successfully.
Student not found.
Time taken to search: 0.000014 seconds.
ok
test_14_delete_nonexistent_student (__main__.TestStudent.test_14_delete_nonexistent_student) ... Student not found.
ok
test_15_average_marks_all_courses (__main__.TestStudent.test_15_average_marks_all_courses) ... Average marks (all courses): 80.06
ok
test_16_median_marks_all_courses (__main__.TestStudent.test_16_median_marks_all_courses) ... Median marks (all courses): 80.00
ok
test_17_average_marks_by_course (__main__.TestStudent.test_17_average_marks_by_course) ... Average marks (course 'DATA200'): 78.09
ok
test_18_stats_nonexistent_course (__main__.TestStudent.test_18_stats_nonexistent_course) ... No numeric marks data found for course 'FAKE999'.
ok
-----
Ran 56 tests in 0.919s
OK

```

I ran the full unit test suite for the CheckMyGrade system using Python's `unittest` framework by executing `test_checkmygrade.py`. The test file programmatically

generates 1000 student records using a loop that creates unique emails (`student0001@test.edu` through `student1000@test.edu`), randomized first and last names, randomly assigned course IDs from the available pool (`DATA200` , `DATA201` , `DATA202` , `DATA203` , `DATA204`), randomly assigned letter grades, and randomly generated numeric marks between 60 and 100 — giving the test suite a large and realistic dataset to work against. The 56 individual test cases cover all major modules including student management, course management, professor management, login/authentication, and grade management. Core CRUD operations are tested such as adding, updating, searching, sorting, and deleting records, alongside edge cases like attempting to update or delete nonexistent students — for example `test_12_update_nonexistent_student` and `test_14_delete_nonexistent_student` — verifying that the system correctly returns "Student not found" rather than crashing. Statistical functions are also validated, including average and median marks computed both globally across all courses and filtered by a specific course, with `test_18_stats_nonexistent_course` confirming that querying a nonexistent course like `FAKE999` gracefully returns "No numeric marks data found" instead of raising an exception. The screenshot also shows the sort and search timing outputs functioning correctly within the test environment — the sort completed in 0.000044 seconds across the full 1000-record dataset and the search resolved in 0.000000 seconds — demonstrating that both features remain highly performant even at scale. The final output shows `Ran 56 tests in 0.919s` with an `OK` status, confirming that every test case passed successfully with no failures or errors.

Links

UML Diagram:

<https://drive.google.com/file/d/17YkvOI0JBCRGFx6Uhhbghkp8Jz09qGSJ/view?usp=sharing>

GitHub Repo: <https://github.com/theonlyjaypas/checkmygrade>